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Catheter assembly with angle changing and related methods

Abstract

A catheter assembly may include a catheter adapter, which may include a distal end, a proximal end, and a lumen extending through the distal end and the proximal end. The catheter assembly may include a catheter secured within the catheter adapter and extending from the distal end of the catheter adapter. The catheter adapter may be configured to move between a first position and a second position. In response to the catheter adapter moving from the first position to the second position, an angle of the catheter adapter with respect to an insertion surface may be configured to increase, which may improve patency of the catheter assembly and alignment of a fluid path of the catheter assembly with vasculature of a patient.

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Background/Summary

(1) The present application claims priority to U.S. Application Ser. No. 63/078,638, entitled “Catheter Assembly With Angle Changing and Related Methods” filed Sep. 15, 2020, the entire disclosure of which is hereby incorporated by reference intis entirety.

BACKGROUND

(1) A catheter is commonly used to infuse fluids into vasculature of a patient. For example, the catheter may be used for infusing normal saline solution, various medicaments, or total parenteral nutrition. The catheter may also be used for withdrawing blood from the patient.

(2) The catheter may include an over-the-needle peripheral intravenous (“IV”) catheter. In this case, the catheter may be mounted over an introducer needle having a sharp distal tip. The catheter and the introducer needle may be assembled so that the distal tip of the introducer needle extends beyond the distal tip of the catheter with the bevel of the needle facing up away from skin of the patient. The catheter and the introducer needle are generally inserted at a shallow angle through the skin into vasculature of the patient.

(3) In order to verify proper placement of the introducer needle and/or the catheter in the blood vessel, a clinician generally confirms that there is “flashback” of blood in a flashback chamber of the catheter assembly. Once placement of the needle has been confirmed, the clinician may remove

the needle, leaving the catheter in place for future blood withdrawal or fluid infusion.
(4) Blood withdrawal using the catheter may be difficult for several reasons, particularly when a dwell time of the catheter within the vasculature is more than one day. When the catheter is left inserted in the patient for a prolonged period of time, the catheter or vein may be more susceptible to narrowing, collapse, kinking, blockage by debris (e.g., fibrin or platelet clots), and adhering of a tip of the catheter to the vasculature. Due to this, the catheter may become compromised for infusion, blood draw, or aspiration over time. The catheter is often used for acquiring a blood sample at a time of catheter placement, but the catheter is less frequently used for acquiring a blood sample during the catheter dwell period. Therefore, when a blood sample is required, an additional needle stick is often used to provide vein access for blood collection, which may be painful for the patient and result in higher material costs.
(5) The subject matter claimed herein is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one example technology area where some implementations described herein may be practiced.
SUMMARY
(6) The present disclosure relates generally to vascular access systems and related devices and methods. In some embodiments, a catheter assembly may be configured to provide angle changing of a catheter, which may reduce a likelihood of occlusion of the catheter, improve patency, and facilitate alignment of a fluid path of the catheter assembly with vasculature of a patient.
(7) In some embodiments, the catheter assembly may include a catheter adapter, which may include a distal end, a proximal end, and a lumen extending through the distal end of the catheter adapter and the proximal end of the catheter adapter. In some embodiments, the catheter assembly may include the catheter secured within the catheter adapter and extending from the distal end of the catheter adapter.
(8) In some embodiments, the catheter assembly may include an expandable support coupled to the catheter adapter. In some embodiments, the expandable support may be configured to adjust an angle of the catheter adapter with respect to skin of the patient. In some embodiments, the expandable support may be configured to move from a diminished position to an expanded position in response to an infusion of liquid or air into the expandable support. In some embodiments, in response to the expandable support being moved from the diminished position to the expanded position, an angle of the catheter adapter with respect to an insertion surface may be configured to increase.
(9) In some embodiments, the expandable support may include a telescoping member. In some embodiments, the expandable support may be configured to move from the diminished position to the expanded position in response to extension of the telescoping member. In some embodiments, in response to the expandable support being moved from the diminished position to the expanded position, an angle of the catheter adapter with respect to an insertion surface may be configured to increase.
(10) In some embodiments, a protrusion may be coupled to the catheter adapter. In some embodiments, the protrusion may be configured to rotate. In some embodiments, in response to the protrusion being rotated from a first position to a second position, an angle of the catheter adapter with respect to an insertion surface may be configured to increase. In some embodiments, in response to the protrusion being rotated from the first position to the second position, the catheter adapter rotates with the protrusion.
(11) In some embodiments, the catheter assembly may include a platform. In some embodiments, in response to the protrusion being in the first position, the catheter adapter may rest on the platform. In some embodiments, in response to the protrusion being rotated to the second position, the protrusion may rest on the platform, and the catheter adapter may be spaced apart from the platform. In these and other embodiments, the catheter adapter and the protrusion may be

monolithically formed as a single unit.
(12) In some embodiments, the catheter assembly may include a strap disposed over the catheter adapter and the protrusion. In some embodiments, a first end and a second end of the strap may be coupled to the platform. In some embodiments, at least a portion of the strap may be elastomeric. In some embodiments, the catheter assembly may include another strap disposed over the catheter adapter and the protrusion. In some embodiments, a first end and a second end of the other strap may be coupled to the platform. In some embodiments, at least a portion of the other strap may be elastomeric. In some embodiments, the catheter adapter may include a side port disposed between the distal end of the catheter adapter and the proximal end of the catheter adapter. In some embodiments, the side port may be disposed between the strap and the other strap.
(13) In some embodiments, the catheter assembly may include a collar that rotates with respect to the catheter adapter. In some embodiments, the collar may include the protrusion. In some embodiments, the collar may be asymmetric.
(14) In some embodiments, the catheter adapter may be configured to move between a particular first position and a particular second position. In some embodiments, in response to the catheter adapter moving from the particular first position to the particular second position, an angle of the catheter adapter with respect to an insertion surface may be configured to increase.
(15) In some embodiments, the catheter assembly may include a platform that is pivotally coupled to the catheter adapter. In some embodiments, the catheter adapter may be configured to pivot between the particular first position and the particular second position or any number of positions in between the first position and the second position. In some embodiments, in response to the catheter adapter pivoting from the particular first position to the particular second position, an angle of the catheter adapter with respect to the insertion surface is configured to increase.
(16) In some embodiments, the catheter assembly may include a platform coupled to the catheter adapter. In some embodiments, the platform may include a first planar surface and a second planar surface proximate the first planar surface. In some embodiments, the first planar surface may be angled with respect to the second planar surface. In some embodiments, the catheter adapter is configured to rock between the particular first position to the particular second position. In some embodiments, in response to the catheter adapter rocking from the particular first position to the particular second position, the angle of the catheter adapter with respect to the insertion surface is configured to increase.
(17) In some embodiments, the catheter adapter may include a slot. In some embodiments, the catheter assembly may include a slider extending through the slot and configured to move between a proximal position and a distal position. In some embodiments, in response to the slider moving from the proximal position to the distal position, the slider may push on the distal end of the catheter adapter to increase the angle of the catheter adapter with respect to the insertion surface.
(18) In some embodiments, the catheter adapter may include a wing and a wall extending generally perpendicular to the wing. In some embodiments, the slot is disposed within the wall. In some embodiments, the slider may be arched. In some embodiments, the slider may include a push tab.
(19) In some embodiments, the catheter assembly may include a catheter adjuster, which may include a distal end, a proximal end, and the catheter extending there through. In some embodiments, the catheter adjuster may be coupled to the distal end of the catheter adapter at a ball joint or a hinge. In some embodiments, in response to the catheter adjuster moving from another particular first position to another particular second position via the ball joint or the hinge, an angle of the catheter with respect to an insertion surface may be configured to increase. In some embodiments, the catheter assembly may include a collar disposed around the distal end of the catheter adapter. In some embodiments, the collar may be configured to lock the ball joint in the other particular first position or the other particular second position or any number of positions in between the other particular first position and the other particular second position.
(20) It is to be understood that both the foregoing general description and the following detailed

description are examples and explanatory and are not restrictive of the invention, as claimed. It should be understood that the various embodiments are not limited to the arrangements and instrumentality illustrated in the drawings. It should also be understood that the embodiments may be combined, or that other embodiments may be utilized and that structural changes, unless so claimed, may be made without departing from the scope of the various embodiments of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Example embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. **1**A is an upper perspective view of an example catheter assembly, illustrating an example expandable support in an example diminished position, according to some embodiments;

(3) FIG. **1**B is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating the expandable support in an example expanded position, according to some embodiments;

(4) FIG. **1**C is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating an example spring-loaded locking mechanism and the expandable support in the expanded position, according to some embodiments;

(5) FIG. **1**D is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating the spring-loaded locking mechanism and the expandable support in the diminished position, according to some embodiments;

(6) FIG. **1**E is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating an example accordion and the expandable support in the diminished position, according to some embodiments;

(7) FIG. **1**F is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating an example accordion and the expandable support in the expanded position, according to some embodiments;

(8) FIG. **1**G is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating an example threaded extension and the expandable support in the diminished position, according to some embodiments;

(9) FIG. **1**H is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating the threaded extension and the expandable support in the expanded position, according to some embodiments;

(10) FIG. **1**I is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating an example hinge and the expandable support in the diminished position, according to some embodiments;

(11) FIG. **1**J is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating the hinge and the expandable support in the expanded position, according to some embodiments;

(12) FIG. **1**K is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating an example multi-hinge extender and the expandable support in the diminished position, according to some embodiments;

(13) FIG. **1**L is an upper perspective view of the catheter assembly of FIG. **1**A, illustrating the multi-hinge extender and the expandable support in the expanded position, according to some embodiments;

(14) FIG. **1**M is an upper perspective view of another catheter assembly, illustrating an example expandable pocket and example compressible pockets, according to some embodiments;

(15) FIG. **1**N is an upper perspective view of the catheter assembly of FIG. **1**M, illustrating the compressible pockets filled with liquid or gas, according to some embodiments;

(16) FIG. **1**O is an upper perspective view of the catheter assembly of FIG. **1**M, illustrating the expandable pocket filled with liquid or gas, according to some embodiments;
(17) FIG. **1**P is an upper perspective view of another catheter assembly, illustrating an example expandable pocket, according to some embodiments;
(18) FIG. **1**Q is an upper perspective view of the catheter assembly of FIG. **1**P, illustrating the expandable pocket prior to filling with liquid or gas, according to some embodiments;
(19) FIG. **1**R is an upper perspective view of the catheter assembly of FIG. **1**P, illustrating the expandable pocket filled with liquid or gas, according to some embodiments;
(20) FIG. **2**A is an upper perspective view of another catheter assembly, illustrating the catheter assembly in an example first position and an example protrusion, according to some embodiments;
(21) FIG. **2**B is a side view of the catheter assembly of FIG. **2**A, illustrating the catheter assembly in the first position, according to some embodiments;
(22) FIG. **2**C is a side view of the catheter assembly of FIG. **2**A, illustrating the catheter assembly in an example second position, according to some embodiments;
(23) FIG. **3**A is a proximal end view of the catheter assembly of FIG. **2**A, illustrating the catheter assembly in the first position, according to some embodiments;
(24) FIG. **3**B is a proximal end view of the catheter assembly of FIG. **2**A, illustrating the catheter assembly in the second position, according to some embodiments;
(25) FIG. **4**A is an upper perspective view of the catheter assembly of FIG. **2**A, illustrating an example collar, according to some embodiments;
(26) FIG. **4**B is a side view of the catheter assembly of FIG. **2**A, illustrating the collar in an example first position, according to some embodiments;
(27) FIG. **4**C is a proximal end view of the catheter assembly of FIG. **2**A, illustrating the collar in the first position, according to some embodiments;
(28) FIG. **4**D is a cross-sectional view of the catheter assembly of FIG. **2**A, illustrating the collar in the first position, according to some embodiments;
(29) FIG. **4**E is a side view of the catheter assembly of FIG. **2**A, illustrating the collar in an example second position, according to some embodiments;
(30) FIG. **4**F is a proximal end view of the catheter assembly of FIG. **2**A, illustrating the collar in the second position, according to some embodiments;
(31) FIG. **5**A is a proximal end view of the catheter assembly of FIG. **2**A, illustrating another example collar in an example first position, according to some embodiments;
(32) FIG. **5**B is a proximal end view of the catheter assembly of FIG. **2**A, illustrating the other example collar in an example second position, according to some embodiments;
(33) FIG. **6**A is a proximal end view of the catheter assembly of FIG. **2**A, illustrating another example collar in an example first position, according to some embodiments;
(34) FIG. **6**B is a proximal end view of the catheter assembly of FIG. **2**A, illustrating the other collar of FIG. **6**A in an example second position, according to some embodiments;
(35) FIG. **7**A is an upper perspective view of another catheter assembly, according to some embodiments;
(36) FIG. **7**B is a side view of the catheter assembly of FIG. **7**A, illustrating the catheter assembly in an example first position, according to some embodiments;
(37) FIG. **7**C is a side view of the catheter assembly of FIG. **7**A, illustrating the catheter assembly in an example second position, according to some embodiments;
(38) FIG. **8**A is an upper perspective view of another catheter assembly, according to some embodiments;
(39) FIG. **8**B is a side view of the catheter assembly of FIG. **8**A, illustrating the catheter assembly in an example first position, according to some embodiments;
(40) FIG. **8**C is a side view of the catheter assembly of FIG. **8**A, illustrating the catheter assembly in an example second position, according to some embodiments;

(41) FIG. **9**A is a cross-sectional view of another example catheter assembly, illustrating an example catheter adjuster in an example first position, according to some embodiments;
(42) FIG. **9**B is a cross-sectional view of the catheter assembly of FIG. **9**A, illustrating the catheter adjuster in an example second position, according to some embodiments;
(43) FIG. **9**C is a cross-sectional view of the catheter assembly of FIG. **9**A, illustrating the catheter adjuster in a locked position, according to some embodiments;
(44) FIG. **9**D is a side view of the catheter assembly of FIG. **9**A, illustrating an example platform pivotally coupled to the catheter adapter, according to some embodiments;
(45) FIG. **10**A is a side view of another catheter assembly, illustrating an example slider in an example distal position, according to some embodiments; and
(46) FIG. **10**B is a side view of the catheter assembly of FIG. **10**A, illustrating the slider in an example proximal position, according to some embodiments.

DETAILED DESCRIPTION

(47) Referring now to FIGS. **1**A-**1**L, in some embodiments, a catheter assembly **10** may be configured to provide angle changing of a catheter **12**, which may reduce a likelihood of occlusion of the catheter **12**, improve patency, and facilitate alignment of a fluid path of the catheter assembly **10** with vasculature of a patient. In some embodiments, the catheter assembly **10** may also move the catheter **12** radially, distally and/or proximally within the vasculature, which may also reduce the likelihood of occlusion and improve patency. In some embodiments, the angle changing provided by the catheter assembly **10** may facilitate blood draw, fluid delivery, patient or device monitoring, or other clinical needs.
(48) In some embodiments, the catheter assembly **10** may include a catheter adapter **14**, which may include a distal end **16**, a proximal end **18**, and a lumen extending through the distal end **16** and the proximal end **18**. In some embodiments, the catheter assembly **10** may include the catheter **12** secured within the catheter adapter **14** and extending from the distal end **16** of the catheter adapter **14**.
(49) In some embodiments, the catheter assembly **10** may include an expandable support **20** coupled to the catheter adapter **14**. In some embodiments, the expandable support **20** may be configured to adjust an angle of the catheter adapter **14** with respect to skin of the patient.
(50) In some embodiments, the expandable support **20** may include a telescoping member **22**, as illustrated, for example, in FIGS. **1**A-**1**B, or another suitable mechanical device to change a height of the proximal end **18** of the catheter adapter **14**. In some embodiments, the expandable support **20** may be configured to move from a diminished position, illustrated, for example, in FIG. **1**A, to an expanded position, illustrated, for example, in FIG. **1**B in response to extension of the telescoping member **22**. In some embodiments, the telescoping member **22** may be configured to lock in a particular position.
(51) In some embodiments, in response to the expandable support **20** being moved from the diminished position to the expanded position, an angle **24** of the catheter adapter **14** with respect to an insertion surface **26** may be configured to increase. In some embodiments, the angle **24** may be measured between a central axis of the catheter adapter and the insertion surface **26**. In these embodiments, the proximal end **18** of the catheter adapter **14** may be lifted up, which may alter a position of the catheter **12**. In some embodiments, as illustrated, in FIGS. **1**A-**1**B, the insertion surface **26** may include skin of the patient.
(52) In some embodiments, a tab **28** may be coupled to the telescoping member **22**. In some embodiments, a user may grip the tab **28** and move the expandable support **20** from the diminished position to the expanded position. In some embodiments, the expandable support **20** may be configured to move from the diminished position to the expanded position in response to an infusion of liquid or air into the expandable support **20**.
(53) As illustrated in FIGS. **1**C-**1**D, in some embodiments, the telescoping member **22** may be coupled to a spring-loaded locking mechanism. In some embodiments, the spring-loaded locking

mechanism may include a spring coupled to a button and the telescoping member **22**. In some embodiments, in response to raising or lengthening the telescoping member **22**, the button may extend through a hole in the telescoping member **22** to secure to the telescoping member in an extended position.

(54) As illustrated in FIGS. **1**E-**1**F, the expandable support **20** may include another suitable mechanical device, such as an accordion extension **29**, to change the height of the proximal end **18** of the catheter adapter **14**. In some embodiments, the expandable support **20** may be configured to move from the diminished position, illustrated, for example, in FIG. **1**E, to the expanded position, illustrated, for example, in FIG. **1**F in response to expansion of the accordion extension **29**. In some embodiments, in response to the accordion extension **29** being in the expanded position, the accordion extension **29** may be configured to remain in the expanded position until additional force from the user. In some embodiments, the accordion extension **29** may be constructed of plastic or another suitable material.

(55) As illustrated in FIGS. **1**G-**1**H, the expandable support **20** may include another suitable mechanical device, such as a threaded extension **31**, to change the height of the proximal end **18** of the catheter adapter **14**. In some embodiments, the threaded extension **31** may be configured to move from the diminished position, illustrated, for example, in FIG. **1**G, to the expanded position, illustrated, for example, in FIG. **1**H in response to unthreading of an upper portion **33** of the threaded extension **31** with respect to a lower portion **35** of the threaded extension of the accordion extension **29**. In some embodiments, the upper portion **33** may include external threads, and the lower portion **35** may include internal threads. In some embodiments, the catheter adapter **14** may rotate with respect to the upper portion **33**.

(56) As illustrated in FIGS. **1**I-**1**J, the expandable support **20** may include another suitable mechanical device, such as a hinge **39**, to change the height of the proximal end **18** of the catheter adapter **14**. In some embodiments, the expandable support **20** may be configured to move from the diminished position, illustrated, for example, in FIG. **1**I, to the expanded position, illustrated, for example, in FIG. **1**J in response to expansion of the hinge **39**. In some embodiments, a spring may extend between ends of the hinge **39** to provide support.

(57) As illustrated in FIGS. **1**K-**1**L, the expandable support **20** may include another suitable mechanical device, such as a multi-hinge extender **41**, to change the height of the proximal end **18** of the catheter adapter **14**. In some embodiments, the expandable support **20** may be configured to move from the diminished position, illustrated, for example, in FIG. **1**K, to the expanded position, illustrated, for example, in FIG. **1**K in response to opposing hinges of the multi-hinge extender **41** moving closer to each other. In some embodiments, the multi-hinge extender **41** may include the opposing hinges and another pair of opposing hinges, and may form a diamond shape.

(58) Referring now to FIGS. **1**M-**1**O, in some embodiments, the catheter assembly **10** may include one or more compressible pockets **43** and/or an expandable pocket **45** to provide angle changing of the catheter **12** and/or positioning of the catheter **12** within the vasculature. In some embodiments, the compressible pockets **43** may be filled with liquid or gas and connected to the expandable pocket via channels **47**. In some embodiments, in response to compression of the compressible pockets **43**, the liquid or the gas may flow from the compressible pockets **43**, through the channels **47**, and into the expandable pocket **45**, which may expand to change the angle of the catheter **12**. In some embodiments, the expandable pocket **45** may be disposed beneath the catheter adapter and/or coupled to the catheter adapter **14** or a platform beneath the catheter adapter **14**. In some embodiments, the expandable pocket **45** may be centrally located between skin of the patient and the catheter adapter **14**. In some embodiments, the compressible pockets **43** may be disposed within one or more wings **51** extending outwardly from the catheter adapter **14**. In some embodiments, there may be a particular compressible pocket **43** within each of the wings **51**, on opposing sides of the catheter adapter **14** for balance.

(59) In some embodiments, in response to a desire to reduce a curve of the catheter **12** to reduce

kinking for infusion or blood draw or when there is an occlusion, the user may press on the wings **51** and compress the compressible pockets **43** to inflate the expandable pocket **45** and raise the catheter adapter **14**.

(60) Referring now to FIGS. **1**P-**1**R, in some embodiments, the expandable pocket **45** may be in fluid communication with an infusion port **53** of the catheter adapter **14**. In some embodiments, the expandable pocket **45** may be filled in response to fluid or gas flowing into the expandable pocket **45** via the infusion port **53**. In some embodiments, the fluid may include saline and/or may be infused into the expandable pocket **45** via a saline flush syringe coupled to an extension set extending from the infusion port **53**. Referring now to FIGS. **2**A-**2**C, a catheter assembly **30** is illustrated, according to some embodiments. In some embodiments, the catheter assembly **30** may be similar or identical to the catheter assembly **10** in terms of one or more included features and/or operation. In some embodiments, a protrusion **32** may be coupled to the catheter adapter **14**. In some embodiments, the protrusion **32** may include an elongated rib. In these and other embodiments, the protrusion **32** may include a flat or planar surface (illustrated, for example, in FIGS. **2**A and **2**C), which may stabilize the protrusion against the insertion surface **26**. In some embodiments, the protrusion **32** may extend along all or a portion of the catheter adapter **14**.

(61) In some embodiments, the protrusion **32** may be configured to rotate. In some embodiments, in response to the protrusion **32** being rotated from a first position to a second position, the angle **24** of the catheter adapter **14** with respect to the insertion surface **26** may be configured to increase. In some embodiments, in response to the protrusion **32** being rotated from the first position to the second position, the catheter adapter **14** may rotate with the protrusion **32**. In these and other embodiments, the protrusion **32** may be fixedly attached to the catheter adapter **14** or the catheter adapter **14** and the protrusion **32** may be monolithically formed as a single unit.

(62) In some embodiments, the catheter assembly **30** may include a platform **34**. In some embodiments, in response to the protrusion **32** being in the first position, illustrated, for example, in FIGS. **2**A-**2**B, the catheter adapter **14** may rest on the platform **34**. In some embodiments, in response to the protrusion **32** being rotated to the second position, illustrated, for example, in FIG. **2**B, the protrusion **32** may rest on the platform **34**, and the catheter adapter **14** may be spaced apart from the platform **34**. In some embodiments, the platform **34** may include securement wings. In some embodiments, the platform **34** may rest on the insertion surface **26**.

(63) In some embodiments, the catheter assembly **30** may include one or more straps **36** disposed over the catheter adapter **14** and the protrusion **32**. In some embodiments, at least a portion of the straps **36** are elastomeric, which may facilitate lifting of the proximal end **18** of the catheter adapter **14**. In some embodiments, the straps **36** may include an elastomeric portion **37**. In some embodiments, an upper portion of the straps **36** may be elastomeric, while a lower portion of the straps **36** may be constructed of a more rigid material to provide support to the catheter adapter **14**.

(64) In some embodiments, the catheter adapter **14** may include side port **38** disposed between the distal end **16** of the catheter adapter **14** and the proximal end **18** of the catheter adapter **14**. In some embodiments, extension tubing may extend from the side port **38** and may be used for infusion into the vasculature and/or blood collection from the vasculature. In some embodiments, the side port **38** may be disposed between two of the straps **36**.

(65) Referring now to FIGS. **3**A-**3**B, the catheter assembly **30** is illustrated, according to some embodiments. In some embodiments, the protrusion **32** may be rounded, as illustrated, for example, in FIGS. **3**A-**3**B. In some embodiments, a first end **40** and a second end **42** of one or more of the straps **36** may be coupled to the platform **34**. In some embodiments, in response to the protrusion **32** being rotated from a first position, illustrated, for example, in FIG. **3**A, to a second position, illustrated, for example, in FIG. **3**B, the angle **24** of the catheter adapter **14** with respect to the insertion surface **26** may be configured to increase. In these embodiments, the proximal end **18** of the catheter adapter **14** may be lifted up, which may alter a position of the catheter **12** and improve patency and/or alignment of a fluid path of the catheter assembly **10** with the vasculature.

(66) Referring now to FIGS. **4**A-**4**F, the catheter assembly **30** is illustrated, according to some embodiments. In some embodiments, the catheter assembly **30** may include a collar **44** that rotates with respect to the catheter adapter **14**. In some embodiments, the collar **44** may be threaded onto the catheter adapter **14**, which may provide secure positioning of the catheter adapter **14**. In some embodiments, the collar **44** may be engaged in a slip-fit with the catheter adapter **14** or coupled to the catheter adapter **14** via another suitable means. In some embodiments, the collar **44** may include the protrusion **32**. In some embodiments, the collar **44** may be asymmetric. In some embodiments, the catheter adapter **14** may include one or more wings, which may be lifted or lowered with the catheter adapter **14**.

(67) In some embodiments, in response to the protrusion **32** being rotated from a first position, illustrated, for example, in FIGS. **4**A-**4**D, to a second position, illustrated, for example, in FIGS. **4**E-**4**F, the angle **24** of the catheter adapter **14** with respect to the insertion surface **26** may be configured to increase. In these embodiments, the proximal end **18** of the catheter adapter **14** may be lifted up, which may alter a position of the catheter **12** and improve patency and/or alignment of a fluid path of the catheter assembly **30** with the vasculature. FIG. **4**D illustrates the lumen **46** of the catheter adapter **14**, which may include a septum **48** to prevent blood leakage through the proximal end **18**.

(68) Referring now to FIG. **5**A-**5**B, the catheter assembly **30** is illustrated, according to some embodiments. In some embodiments, the collar **44** may include various shapes, such as a smooth curve **49** proximate a radial surface **50**, which may be generally perpendicular to the smooth curve **49** and may facilitate rotation by the user. In some embodiments, the collar **84** may entirely surround the catheter adapter **14**. In some embodiments, in response to the protrusion **32** being rotated from the first position, illustrated, for example, in FIG. **5**A, to the second position, illustrated, for example, in FIG. **5**B, an angle (see, for example, the angle **24** of FIGS. **1**A-**1**B of FIGS. **2**B-**2**C) of the catheter adapter **14** with respect to the insertion surface **26** may be configured to increase.

(69) Referring now to FIG. **6**A-**6**B, the catheter assembly **30** is illustrated, according to some embodiments. In some embodiments, the collar **44** may include various shapes, such as a smooth curve **52** proximate a tab **54** to facilitate gripping and rotation by the user. In some embodiments, in response to the protrusion **32** being rotated from the first position, illustrated, for example, in FIG. **5**A, to the second position, illustrated, for example, in FIG. **5**B, an angle of the catheter adapter **14** with respect to the insertion surface **26** may be configured to increase.

(70) Referring now to FIGS. **7**A-**7**C, a catheter assembly **56** is illustrated, according to some embodiments. In some embodiments, the catheter assembly **56** may be similar or identical to the catheter assembly **10** and/or the catheter assembly **30** in terms of one or more included features and/or operation.

(71) In some embodiments, the catheter assembly **56** may include a platform **58** that is pivotally coupled to the catheter adapter **14**. In some embodiments, the catheter adapter **14** may be configured to pivot between a first position, illustrated, for example, in FIGS. **7**A-**7**B, and a second position, illustrated, for example, in FIG. **7**C. In some embodiments, in response to the catheter adapter **14** pivoting from the first position to the second position, an angle of the catheter adapter **14** with respect to the insertion surface is configured to increase, and the proximal end **18** of the catheter adapter **14** may be lifted. In some embodiments, a first shaft **60***a* and a second shaft **60***b* may extend from opposite sides of the catheter adapter **14** and may be configured to rotate within grooves **62***a*, **62***b*, respectively, to move the catheter adapter **14** between the first position and the second position. In some embodiments, the grooves **62***a*, **62***b* may be disposed within walls **64** extending upwardly from the platform **58**.

(72) Referring now to FIGS. **8**A-**8**C, a catheter assembly **66** is illustrated, according to some embodiments. In some embodiments, the catheter assembly **66** may be similar or identical to one or more of the following: the catheter assembly **10**, the catheter assembly **30**, and the catheter

assembly **56**, in terms of one or more included features and/or operation.
(73) In some embodiments, the catheter assembly **66** may include a platform **68** coupled to the catheter adapter **14**. In some embodiments, the platform **68** may include a first planar surface **70** and a second planar surface **72** proximate the first planar surface **70**. In some embodiments, the first planar surface **70** may be angled with respect to the second planar surface **72**. In some embodiments, the catheter adapter **14** may be configured to rock between a first position, illustrated, for example, in FIGS. **8**A-**8**B, to a second position, illustrated, for example, in FIG. **8**C. In some embodiments, the first planar surface **70** may contact the insertion surface **26** in the first position, and the second planar surface **72** may be spaced apart from the insertion surface **26**. In some embodiments, the second planar surface **72** may contact the insertion surface **26** in the second position, and the first planar surface **70** may be spaced apart from the insertion surface **26**.
(74) In some embodiments, in response to the user rocking the catheter adapter **14** from the first position to the second position, the angle **24** of the catheter adapter **14** with respect to the insertion surface may be configured to increase, and the proximal end **18** of the catheter adapter **14** is lifted up, changing a position of the catheter **12** and improving patency and alignment.
(75) Referring now to FIGS. **9**A-**9**C, a catheter assembly **74** is illustrated, according to some embodiments. In some embodiments, the catheter assembly **74** may be similar or identical to one or more of the following: the catheter assembly **10**, the catheter assembly **30**, the catheter assembly **56**, and the catheter assembly **66**, in terms of one or more included features and/or operation.
(76) In some embodiments, the catheter assembly **74** may include a catheter adjuster **76**, which may include a distal end **78**, a proximal end **80**, and the catheter **12** extending there through. In some embodiments, the catheter adjuster **76** may be coupled to the distal end **16** of the catheter adapter **14** at a ball joint **82**, a hinge, or another suitable mechanism.
(77) As illustrated, for example, in FIGS. **9**A-**9**C, the distal end **16** of the catheter adapter **14** may include a socket and the proximal end **80** of the catheter adjuster **76** may include a ball or rounded portion configured to rotate within the socket. Alternatively, in some embodiments, the distal end **16** of the catheter adapter **14** may include the ball or rounded portion and the proximal end **80** of the catheter adjuster **76** may include the socket.
(78) In some embodiments, the catheter adjuster **76** may move from a first position, illustrated, for example, in FIG. **9**A, to a second position, illustrated, for example, in FIG. **9**B, via the ball joint **82** or the hinge. In some embodiments, in response to the catheter adjuster **76** moving from the first position to the second position, an angle of the catheter **12** with respect to the insertion surface **26** may be configured to increase.
(79) As illustrated, for example, in FIG. **9**C, the catheter assembly **74** may include a collar **84** disposed around the distal end **16** of the catheter adapter **14**. In some embodiments, the collar **84** may be configured to lock the ball joint **82** in the first position or the second position. In some embodiments, the collar **84** may include threads corresponding to other threads of an outer surface of the distal end **16** of the catheter adapter **14**. In some embodiments, the collar **84** may partially or entirely surround the catheter adapter **14**.
(80) Referring now to FIG. **9**D, in some embodiments, the catheter assembly **74** may include the platform **58**, which may be pivotally coupled to the catheter adjuster **76** through which the catheter **12** extends. In some embodiments, the platform **58** may include a protrusion **59** upon which the catheter adapter **14** may rest.
(81) In some embodiments, the catheter adjuster **76** may be configured to pivot between the first position and the second position via the hinge. In some embodiments, in response to the catheter adjuster **76** pivoting from the first position to the second position, the catheter **12** may bend and an angle of a distal end of the catheter **12** with respect to the insertion surface is configured to increase. In some embodiments, the proximal end **18** of the catheter adapter **14** may remain still on the protrusion **59** or the platform **58** in response to the catheter adjuster **76** pivoting from the first position to the second position. In some embodiments, the first shaft **60***a* and the second shaft **60***b*

may be fixed to the catheter adjuster **76**. In some embodiments, the first shaft **60***a* and the second shaft **60***b* may extend from opposite sides of the catheter adapter **14** and may be configured to rotate within grooves **62***a*, **62***b* (see, for example, FIG. **7**A), respectively, to move the catheter adjuster **76** and the catheter **12** between the first position and the second position. In some embodiments, the grooves **62***a*, **62***b* may be disposed within walls **64** extending upwardly from the platform **58**.

(82) Referring now to FIGS. **10**A-**10**B, a catheter assembly **86** is illustrated, according to some embodiments. In some embodiments, the catheter assembly **86** may be similar or identical to one or more of the following: the catheter assembly **10**, the catheter assembly **30**, the catheter assembly **56**, the catheter assembly **66**, and the catheter assembly **74**, in terms of one or more included features and/or operation.

(83) In some embodiments, the catheter adapter **14** may include a slot **88**. In some embodiments, the catheter assembly **86** may include a slider **90** extending through the slot **88** and configured to move within the slot **88** between a proximal position, illustrated, for example, in FIG. **10**B, and a distal position, illustrated, for example, in FIG. **10**A. In some embodiments, in response to the slider **90** moving from the proximal position to the distal position, the slider may push on the distal end **16** of the catheter adapter **14** to increase the angle of the catheter adapter **14** with respect to the insertion surface **26**.

(84) In some embodiments, the catheter adapter **14** may include a wing **92** and a wall **94** extending generally perpendicular to the wing **92**. In some embodiments, the slot **88** is disposed within the wall **94**. In some embodiments, the slider **90** may be arched. In some embodiments, the slider **90** may include a protrusion or push tab **96** disposed on a top of the slider **90** and configured for gripping by the user. In some embodiments, the catheter adapter **14** may not include the slide **90**, the wall **94**, and the slot **88**, which may be incorporated into a stabilization device or dressing coupled to the catheter adapter **14** or near the catheter adapter **14**.

(85) All examples and conditional language recited herein are intended for pedagogical objects to aid the reader in understanding the invention and the concepts contributed by the inventor to furthering the art and are to be construed as being without limitation to such specifically recited examples and conditions. Although embodiments of the present inventions have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A catheter assembly, comprising: a catheter adapter, comprising a distal end, a proximal end, and a lumen extending through the distal end and the proximal end; a catheter secured within the catheter adapter and extending from the distal end of the catheter adapter; an expandable support coupled to the catheter adapter, wherein the expandable support has an adjustable height configured to adjust an angle of the catheter adapter with respect to skin of the patient, wherein the adjustable height of the expandable support is configured to increase from a diminished position to an expanded position in response to an infusion of liquid or air into the expandable support.
2. A catheter assembly, comprising: a catheter adapter, comprising a distal end, a proximal end, and a lumen extending through the distal end and the proximal end; a catheter secured within the catheter adapter and extending from the distal end of the catheter adapter; and a protrusion coupled to the catheter adapter, wherein the protrusion is configured to rotate about a central longitudinal axis of the catheter adapter, wherein in response to the protrusion being rotated from a first position to a second position, an angle of the catheter adapter with respect to an insertion surface is configured to increase.
3. The catheter assembly of claim 2, wherein in response to the protrusion being rotated from the first position to the second position, the catheter adapter rotates with the protrusion.

4. The catheter assembly of claim 3, further comprising a platform, wherein in response to the protrusion being in the first position, the catheter adapter rests on the platform, wherein in response to the protrusion being rotated to the second position, the protrusion rests on the platform and the catheter adapter is spaced apart from the platform.
5. The catheter assembly of claim 4, wherein the catheter adapter and the protrusion are monolithically formed as a single unit.
6. The catheter assembly of claim 2, further comprising a strap disposed over the catheter adapter and the protrusion, wherein a first end and a second end of the strap are coupled to the platform, wherein at least a portion of the strap is elastomeric.
7. The catheter assembly of claim 6, further comprising another strap disposed over the catheter adapter and the protrusion, wherein a first end and a second end of the other strap are coupled to the platform, wherein at least a portion of the other strap is elastomeric.
8. The catheter assembly of claim 7, wherein the catheter adapter further comprises a side port disposed between the distal end of the catheter adapter and the proximal end of the catheter adapter, wherein the side port is disposed between the strap and the other strap.
9. The catheter assembly of claim 2, further comprising a collar that rotates with respect to the catheter adapter, wherein the collar comprises the protrusion, wherein the collar is asymmetric.
10. A catheter assembly, comprising: a catheter adapter, comprising a distal end, a proximal end, and a lumen extending through the distal end and the proximal end; a catheter secured within the catheter adapter and extending from the distal end of the catheter adapter; and a platform coupled to the catheter adapter, wherein the platform comprises a first planar surface and a second planar surface proximate the first planar surface, wherein the first planar surface is angled with respect to the second planar surface, wherein the first and second planar surfaces are configured to contact an insertion surface of a patient, wherein the catheter adapter is configured to rock in a direction along a central longitudinal axis of the catheter adapter between (i) a first position in which the first planar surface contacts the insertion surface, to (ii) a second position in which the second planar surface contacts the insertion surface, wherein in response to the catheter adapter rocking from the first position to the second position, the angle of the catheter adapter with respect to the insertion surface is configured to increase.
11. A catheter assembly, comprising: a catheter adapter, comprising a distal end, a proximal end, and a lumen extending through the distal end and the proximal end; and a catheter secured within the catheter adapter and extending from the distal end of the catheter adapter, wherein the catheter adapter comprises a slot, further comprising a slider extending through the slot and configured to move between a proximal position and a distal position, wherein in response to the slider moving from the proximal position to the distal position, the slider pushes on the distal end of the catheter adapter to increase the angle of the catheter adapter with respect to an insertion surface.
12. The catheter assembly of claim 11, wherein the catheter adapter comprises a wing and a wall extending generally perpendicular to the wing, wherein the slot is disposed within the wall.
13. The catheter assembly of claim 11, wherein the slider is arched.
14. The catheter assembly of claim 11, wherein the slider comprises a push tab on a top of the slider.
15. A catheter assembly, comprising: a catheter adapter, comprising a concave distal end, a proximal end, and a lumen extending through the distal end and the proximal end; a catheter secured within the catheter adapter and extending from the distal end of the catheter adapter; a catheter adjuster comprising a distal end and a convex proximal end, wherein the proximal end of the catheter adjuster is coupled to the distal end of the catheter adapter to form a ball joint, wherein the catheter extends through the catheter adjuster and the ball joint, wherein in response to the catheter adjuster moving from a first position to a second position via the ball joint, an angle of the catheter with respect to an insertion surface is configured to increase; and a collar threadably

disposed around the distal end of the catheter adapter, wherein the collar is configured to lock the ball joint in the first position or the second position.